Prime subfield residues do not extend to rank four

Human Compression project

Abstract

We complete the three extension arguments in Section 3.1 of the fixed higher-rank Suzuki article. None of the prime split-torus rank-three residues extends to a rank-four family satisfying the source's subgroup axiom. The proof uses maximal subgroups of an odd dihedral group and uniqueness of an embedded affine subgroup inside a fixed-size subfield. In particular, equal abstract orders are never used to identify two subfield groups. The surviving intermediate candidate in the source's second case is excluded by the already proved alignment of split tori.

1 Classical and previously proved inputs

Write $q = 2^n$, with n odd, and $G = \mathrm{Sz}(q)$. For a family (G_i) put $G_J = \bigcap_{j\in J} G_j$. We use exactly the source condition that for every proper J, some unused type gives a maximal proper inclusion $G_{J\cup\{i\}} < G_J$. The acquired strict-intersection lemma implies that adding *any* unused type strictly decreases the intersection. An induced family in any G_J satisfies the same axiom. No flag-transitivity hypothesis is imposed.

The classical Suzuki subgroup classification, torus partition and ovoid model are retained as explicit premises. A nonidentity element in a split torus has that torus as its centralizer. A full split torus has normalizer $D_q = C_{q-1} : C_2$. No point stabilizer contains an odd dihedral group with nontrivial rotation subgroup: in its abelian odd quotient, an odd rotation has nontrivial image, while an inverting involution has trivial image, which is impossible. A proper nonsolvable subgroup is a simple subfield group. A solvable subgroup is contained in a point stabilizer or one of the torus normalizers. These are the subgroup-classification consequences already used in the accepted common notes.

We also reuse the following proved consequences.

1. Every parabolic in an odd-dihedral family satisfying the subgroup axiom is maximal in that dihedral group.

2. The full split normalizer of a standard $\mathrm{Sz}(s)$ is self-normalizing in G, and determines exactly one embedded subfield of a prescribed containing degree. The same-size affine subgroup $A_s = Z_s : C_{s-1}$ determines exactly one copy of $\mathrm{Sz}(s)$.

3. The proper overgroups of a full split torus in $\mathrm{Sz}(t)$ are that torus, its split normalizer, the two affine groups at its fixed points, and the two point stabilizers. A point stabilizer cannot be a parabolic of a family of rank at least two satisfying the subgroup axiom.

4. If $t = 2^f$ has prime degree f but composite $t - 1$, there is no family of rank at least three satisfying the subgroup axiom.

5. Two full split dihedrals of subfield groups inside a common split normalizer, sharing an involution, align the two subfields. Their intersection is the standard group over the intersection field.

These are respectively the accepted cyclic/dihedral lemma, embedded normalizer and affine uniqueness lemmas, full-torus overgroup lemma, prime-degree obstruction, and shared-reflection alignment lemma. Their proofs are imported explicitly; the source's exhaustive geometry list is not a premise.

2 One prime-degree maximal subfield

Suppose $G_0 = \mathrm{Sz}(s)$ is maximal in G, where $s = 2^d \geq 8$ and $p = s - 1$ is prime. Then d is an odd prime, and $n = dr$ for an odd prime r. The two primes d, r may coincide. Fix its full split dihedral $\Delta_s = D_{2p}$ and an affine subgroup $A_s = Z_s : C_p$.

Lemma 1. *A proper subgroup containing Δ_s and not contained in G_0 is contained in the unique ambient split normalizer D_q containing Δ_s. A proper subgroup containing A_s and not contained in G_0 is contained in the unique point stabilizer fixed by Z_s.*

Proof. An odd element of order p determines a split torus in G. A simple subfield $\mathrm{Sz}(2^f)$ containing it must have $p \mid 2^f - 1$: its nonsplit orders divide $(2^f)^2 + 1$, which divides $q^2 + 1$, and $q-1$ is coprime to q^2+1. Since $p = 2^d - 1$ is prime, 2 has order d modulo p; it cannot have order one, and d is prime. Thus $d \mid f$. The only proper divisor f of dr that is divisible by d is $f = d$. The acquired embedded uniqueness for Δ_s or A_s therefore makes such a containing subfield equal to G_0.

A subgroup containing Δ_s cannot lie in a point stabilizer, by the odd-dihedral observation, or in a nonsplit normalizer, by torus coprimality. Its only remaining soluble position is in the unique D_q fixed by its rotation subgroup. A subgroup containing A_s cannot lie in any torus normalizer: it contains the elementary abelian Z_s of order $s \geq 8$, whereas their Sylow two-subgroups have order two or four. Its point stabilizer is unique, since each nonidentity element of Z_s fixes precisely one ovoid point. The subgroup classification now proves both assertions. □

Lemma 2. *Suppose a rank-four family has $G_{01} = \Delta_s$, and $G_{013} \cong C_2$. Then*

$$G_1 = D_{2p\ell}, \qquad G_{13} = D_{2\ell}, \tag{1}$$

where $\ell \neq p$ is an odd prime dividing $(q-1)/p$. If $G_{012} \cong C_2$, then also $G_{12} = D_{2\ell}$ and $G_{123} = C_\ell$. If $G_{012} = C_p$, then G_{12} is either $C_{p\ell}$ or a dihedral subgroup of order $2p$ distinct from G_{01}.

Proof. Strict intersections give $G_1 \not\leq G_0$. Lemma 1 places it inside D_q. It contains Δ_s, so is dihedral. The induced rank-three family in G_1 has all parabolics maximal. Maximality of Δ_s gives $G_1 = D_{2p\ell}$ with ℓ prime. If $\ell = p$, every maximal subgroup of G_1 contains its order-p rotation subgroup, contradicting $G_{013} \cong C_2$. For distinct p, ℓ, its maximal subgroups are the rotation group $C_{p\ell}$ and dihedrals of orders $2p$ and 2ℓ. Only the last type can intersect G_{01} in order two. A dihedral of order 2ℓ and G_{01} have a common reflection: their reflection congruences in the cyclic rotation group have coprime moduli p, ℓ. Thus the asserted intersection also has that order. Two distinct dihedrals of order 2ℓ in G_1 share exactly their order-ℓ rotation group, proving the last intersection claim. The case of order-p intersection leaves precisely the other two maximal-subgroup types. Equality of any two displayed parabolics is excluded by strict intersections. □

Lemma 3. *Under (1), suppose G_{03} is not contained in G_{01}. Then $G_3 = \mathrm{Sz}(t)$, where $t = 2^r$, $r \neq d$, and $t - 1 = \ell$ is prime.*

Proof. The group G_3 contains $G_{13} = D_{2\ell}$. This excludes a point stabilizer and the nonsplit normalizers. If G_3 were soluble, it would therefore lie in D_q, the unique split normalizer containing this dihedral. But $G_0 \cap D_q = \Delta_s$: its odd rotation subgroup is the full split torus of G_0. This would put G_{03} inside G_{01}, a contradiction.

Thus G_3 is a proper simple subfield group. Its possible field exponents are d and r, since $n = dr$. The order-ℓ rotation is split in this subfield by the same coprimality argument as above. The first size is impossible, since $\ell \neq p = s - 1$. Consequently $G_3 = \mathrm{Sz}(2^r)$ and $r \neq d$. Its induced rank-three family satisfies the subgroup axiom. The prime-degree obstruction implies that $2^r - 1$ is prime. Since it is divisible by ℓ, it equals ℓ. □

3 The three residue types

Theorem 4. *No rank-four family satisfying the source axiom has a maximal parabolic $G_0 = \mathrm{Sz}(s)$ with $s - 1$ prime. In particular none of residues R_1, R_2, R_3 in Section 3.1 of the fixed L110 source extends as proposed.*

Proof. The acquired necessary-pattern theorem gives exactly these three possibilities in the induced rank-three family of G_0. In all cases we can label $G_{01} = \Delta_s$ and $G_{013} = C_2$.

For R_1, we have $G_{02} = A_s$ and $G_{012} = C_2$. Lemma 2 gives $G_{12} = D_{2\ell}$. By strict intersections $G_2 \not\leq G_0$, so Lemma 1 puts G_2 in a point stabilizer. But it contains the odd dihedral G_{12}, a contradiction. This proves the first nonextension directly, without any appeal to an ovoid-circle intersection assumption.

For R_2, $G_{02} = A_s$, $G_{03} = D_{2a}$ for an odd prime a, and $G_{012} = C_p$. Again G_2 lies in its unique point stabilizer, so it cannot contain a dihedral group of order $2p$. Lemma 2 therefore gives $G_{12} = C_{p\ell}$. The subgroup G_{03} is not contained in G_{01} by strict intersections, so Lemma 3 gives $G_3 = \mathrm{Sz}(t)$ with $t - 1 = \ell$. The full split dihedrals G_{01} and G_{13} of G_0, G_3 lie in G_1 and share $G_{013} = C_2$. Alignment forces

$$G_0 \cap G_3 \cong \mathrm{Sz}(2^{\gcd(d,r)}) = \mathrm{Sz}(2),$$

which has order twenty. This contradicts $G_{03} = D_{2a}$ for prime a. The source's final $a = 5$ candidate is therefore impossible as well; the extra affine parabolic cannot repair this obstruction.

For R_3, $G_{02} = D_{2a}$ and $G_{03} = D_{2b}$, where the permitted half-orders are primes and at most one is two. All three intersections $G_{012}, G_{013}, G_{023}$ have order two, and the full intersection is trivial. Lemma 2 gives

$$G_{12}, G_{13} \cong D_{2\ell}, \qquad G_{123} = C_\ell.$$

Apply Lemma 3 also with types 2 and 3 interchanged. Both G_2, G_3 are copies of the same $\mathrm{Sz}(t)$, where $t-1 = \ell$. Equal sizes alone do not identify these subgroups. Instead their common subgroup G_{23} contains the full split torus C_ℓ. Within G_2, the acquired full-torus overgroup list leaves C_ℓ, its normalizer $D_{2\ell}$, the two affine groups, and the two point stabilizers. The first choice would equal G_{123}; the second would equal G_{12}, since the full torus has a unique normalizer. Both violate strict intersections. A point stabilizer is excluded as a parabolic of the induced family in G_2. Thus G_{23} is an affine group $A_t = Z_t : C_{t-1}$. The embedded affine uniqueness theorem now forces its two containing copies G_2, G_3 of $\mathrm{Sz}(t)$ to coincide, the final contradiction. □

Attribution and scope. The source is D. Leemans, *The residually weakly primitive pregeometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, especially printed pp. 10–12. The source's case division and nonextension conclusions for R_1, R_3 are retained; the proofs here make their overgroup and uniqueness steps explicit. The R_2

obstruction reuses the earlier shared-reflection proof already applied to the full prescribed lattice of Construction 1. The general Suzuki structure is classical. No assertion that the other composite-degree residue lists are exhaustive, no whole-original completion credit, and no additional search, Lean result or complexity measurement is made by this component.